

Worksheet 2.1

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Subject Code: 25CAP-656

1. Aim/Overview of the practical: To design and develop a Java EE application that demonstrates inter-servlet communication by using `RequestDispatcher` to forward requests and `sendRedirect()` to redirect responses between servlets.

2. Coding:

Index.html file:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="bg-light d-flex justify-content-center align-items-center vh-100">

  <form class="p-4 shadow rounded bg-white w-45 mx-3" action="Servlet1" method="post">
    <h3 class="text-center mb-4">Registration Form using post method</h3>

    <div class="mb-3">
      <label for="name" class="form-label">Name</label>
      <input type="text" class="form-control" name="name" id="name" placeholder="Enter your name">
    </div>

    <div class="mb-3">
      <label for="age" class="form-label">Age</label>
      <input type="number" class="form-control" name="age" id="age" placeholder="Enter your age">
    </div>

    <button type="submit" class="btn btn-primary w-100">Submit</button>
  </form>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```



Servlet1 file:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/Servlet1"})
public class Servlet1 extends HttpServlet {
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

        String name = request.getParameter("name");
        String age = request.getParameter("age");

        request.setAttribute("userName", name);
        request.setAttribute("userAge", age);

        RequestDispatcher rd = request.getRequestDispatcher("Servlet2");
        rd.forward(request, response);
        // response.sendRedirect("https://github.com/");
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}
```



Servlet2 file:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/Servlet2"})
public class Servlet2 extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

        String name = (String) request.getAttribute("userName");
        String age = (String) request.getAttribute("userAge");

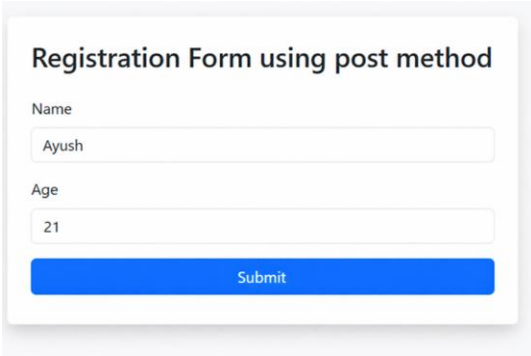
        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<body>");
            out.println("Name: " + name + "<br>");
            out.println("Age: " + age);
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

Servlet1 with sendRedirect() method:

```
RequestDispatcher rd = request.getRequestDispatcher("Servlet2");
response.sendRedirect("https://github.com/");
```

----- rest all code same in servlet1 -----

OUTPUT:



Registration Form using post method

Name
Ayush

Age
21

Submit

Figure 1: index.html file

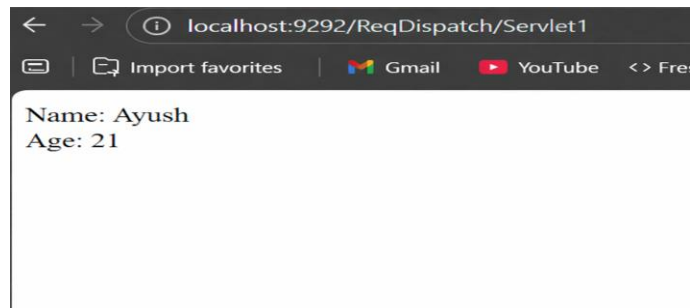


Figure 2: with forward() method

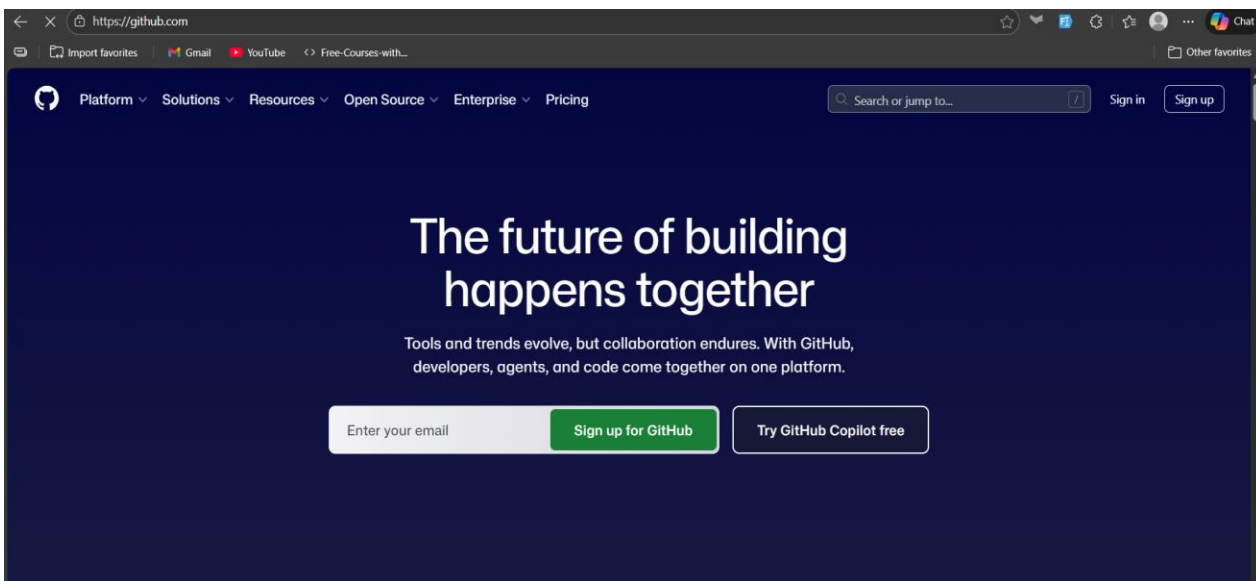


Figure 3: with sendRedirect() method

3. Learning outcomes:

- Understood how to transfer control between servlets using RequestDispatcher.
- Learned the difference between forward() and sendRedirect() in servlets.
- Implemented data sharing between servlets using request attributes.
- Observed how request scope data behaves during forwarding and redirection.
- Learned how form data flows from HTML to multiple servlets.